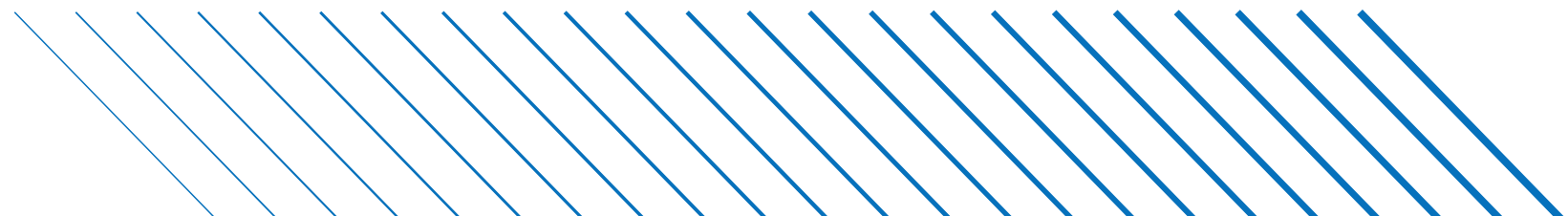
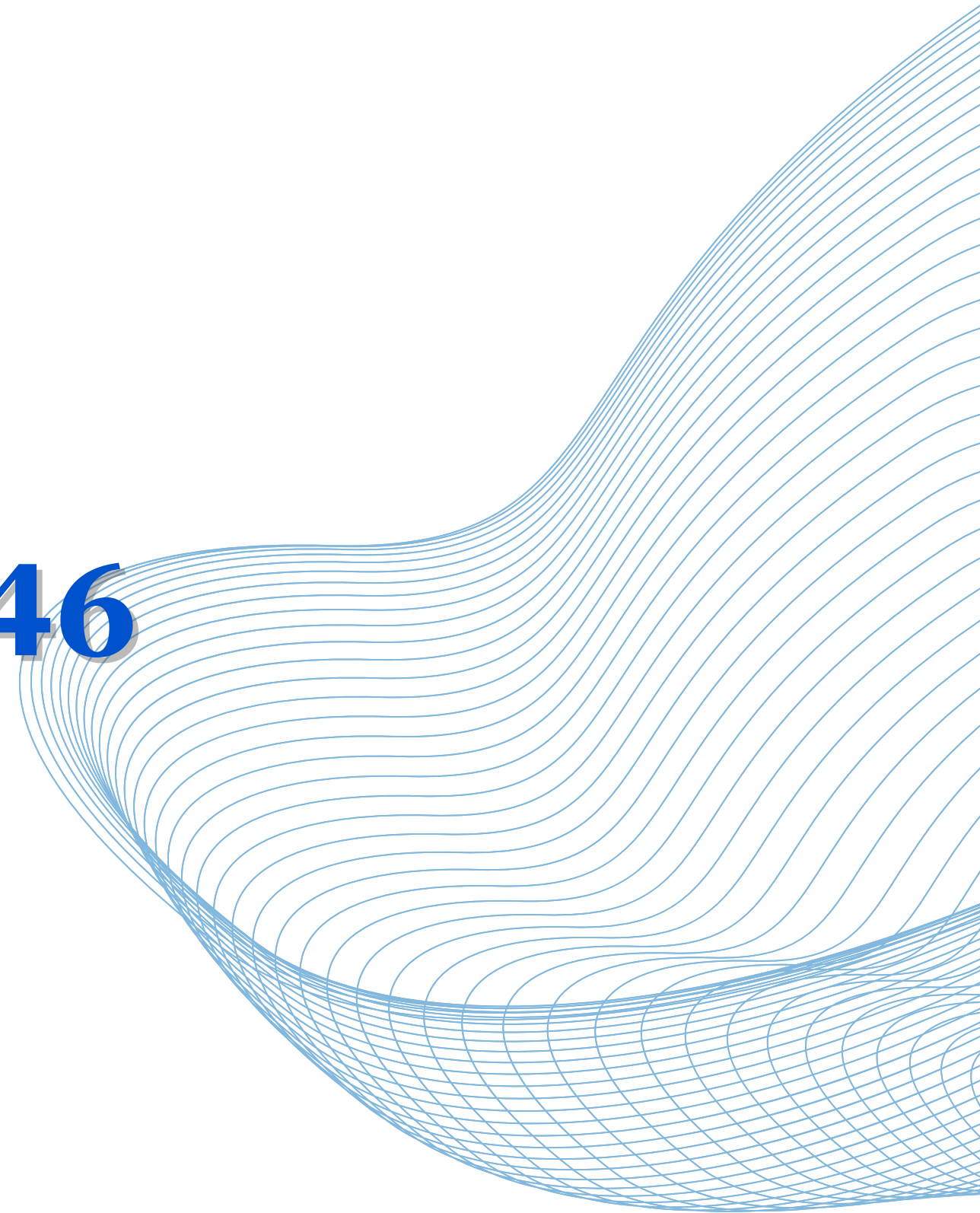


SageArk

Quant Training Session 45-46

Risk of a Portfolio and Optimization



What is Risk Management?

- **Process by which we keep “how much did we lose” strictly less than “how much should we lose?” and “how much could we lose?”**
 - Practiced by both traders and risk managers
 - Blends market understanding with quantitative models
 - Clearly defined intervention points in adverse situations
- **Scope can be different**
 - Desks look at their risk
 - Trading managers look at the groups they oversee
 - Risk Management and Partners aggregate everyone

Big Losses

**Archegos
(2021)**

Long stocks through
swap lines - \$20B

**Melvin
Capital
(2021)**

Short GameStop
- \$7B

Parplus

Vix
Replication -
\$500mm+

**Einar
Aas/Nasdaq
(2018)**

German vs
Scandinavian
Power - \$125mm

**LTCM
(1998)**

Bond RV -
\$4B

**Amaranth/
Brian Hunter
(2006)**

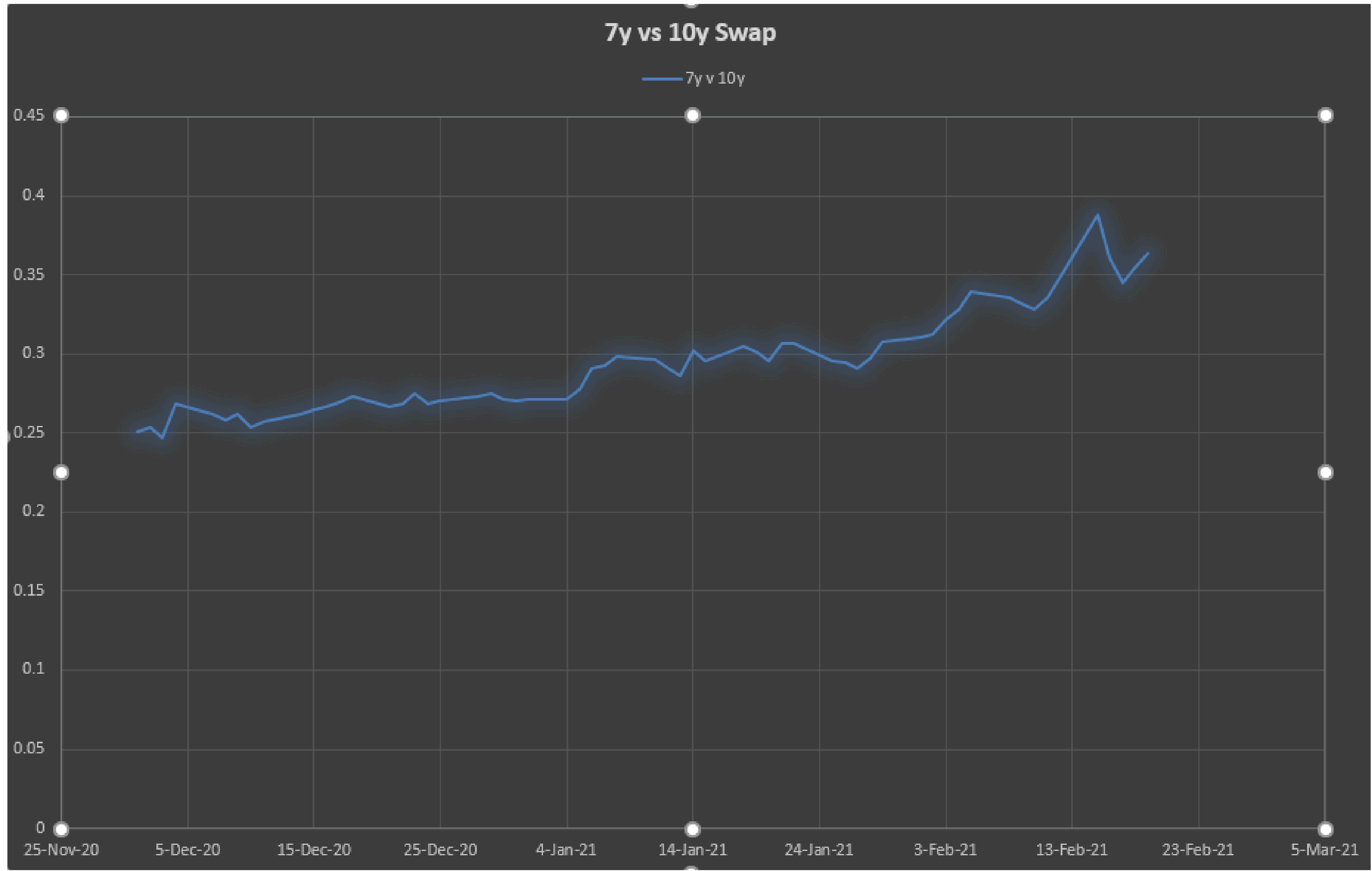
Natural Gas
Futures - \$6.5B

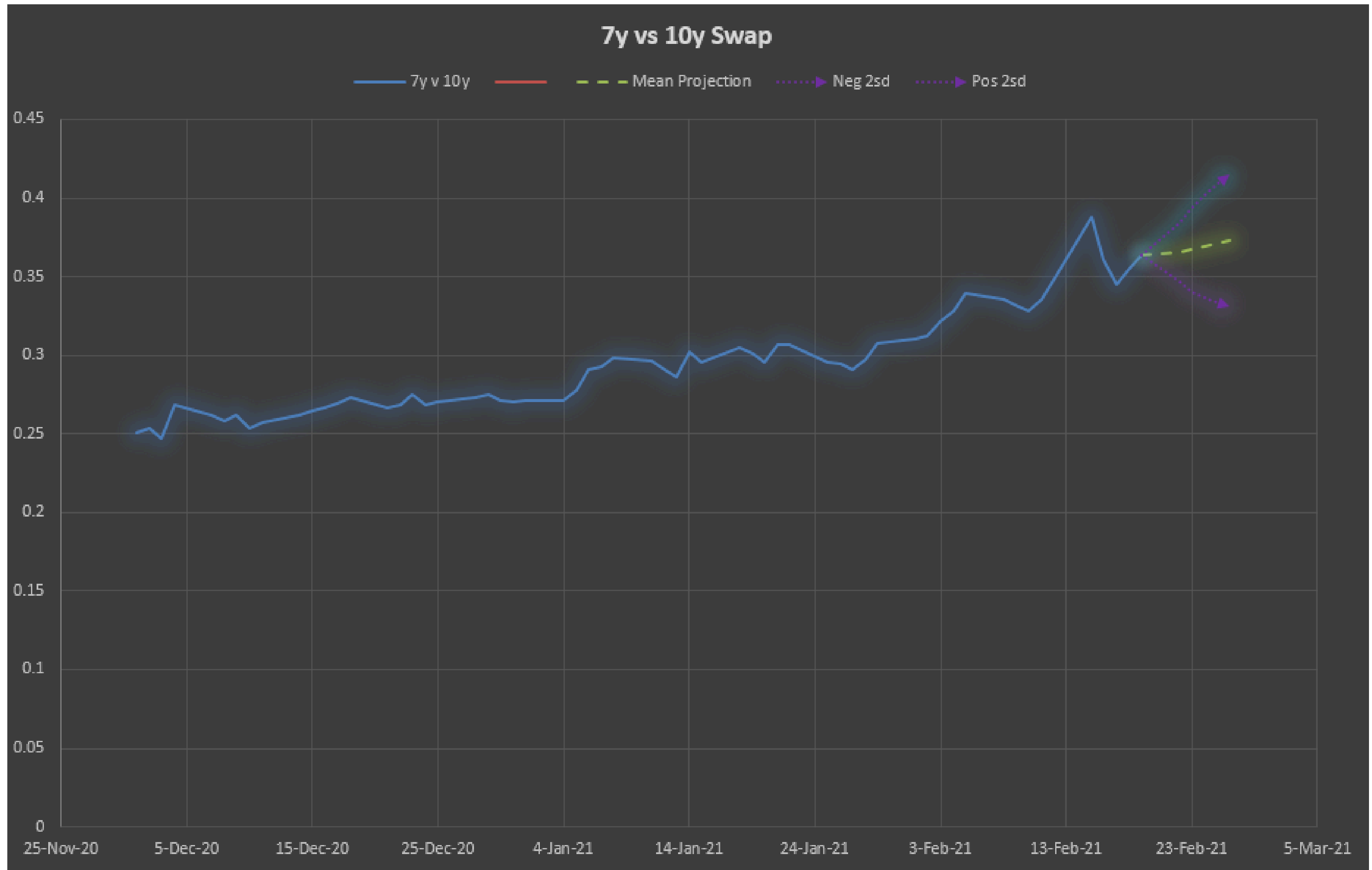
Assumptions

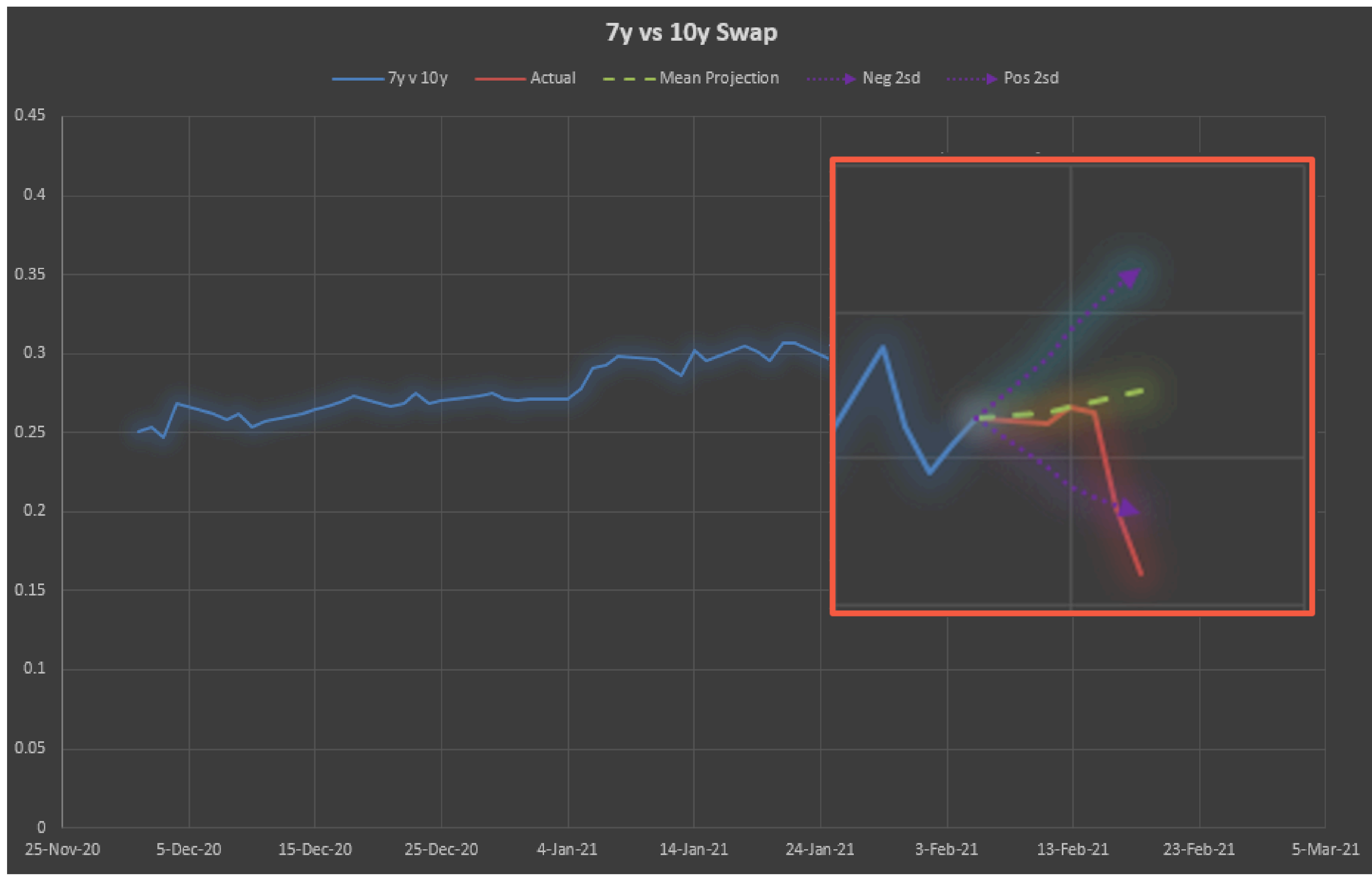
What are some standard assumptions we make in financial modeling?

Exercise:

- “Swap7s10s.csv” contains observations of the 7y vs 10y swap rates from the start of the year through mid February
- Predict the next 5 days of data, and draw the +/- 2sd confidence levels







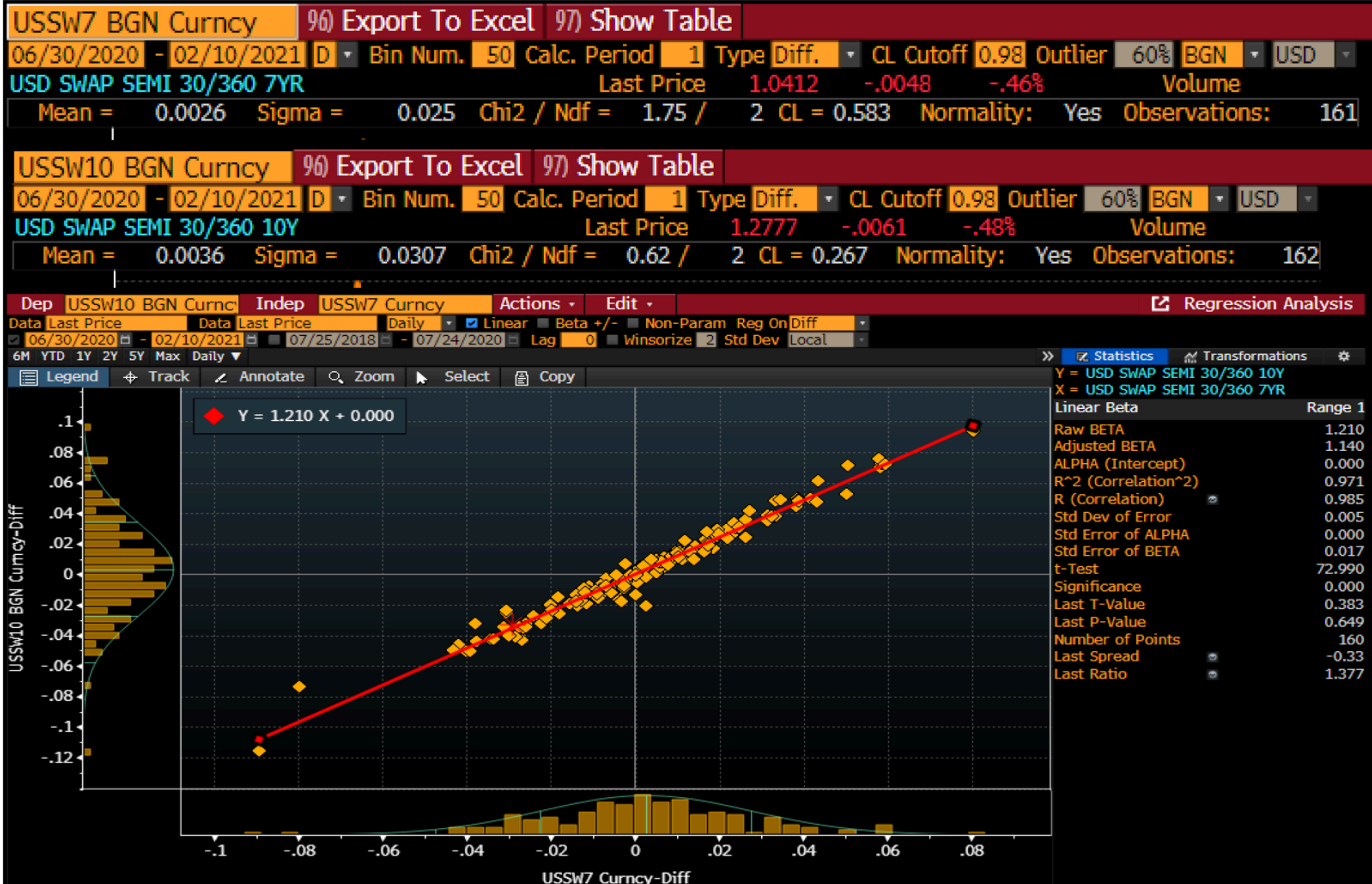
Bayesian Analysis (kind of)

See an opportunity. Spend time researching.

- Four “P”s of risk taking
 - Patience
 - Preparation
 - Paranoia
 - Proactive
- Execute trade, as desired
 - Market performs out of sample/out of expectation
- WHAT NOW?

Initial risk management is very important.

- Size trade appropriately so that the current MTM loss is manageable
 - If not, too bad. You’re out of the risk whether you like it or not
- Re-evaluate. Is this trade what you thought it was?
 - Don’t be sore, add some more
 - Cut your losses and fight another day



Example Trade

- Think 7s/10s can continue same trend over next 3 months
- Estimate another 10bp return, daily SD ~ 0.5 bp
 - Correlation of .985, vol of 3bp on 10y and 2.5bp on 7y
- Buy 10y rate to max size when spread deviates 3bp from trend
 - Buy \$250k/bp to start, add \$250k/bp per 1bp dislocation
 - Down \$1.5mm at max size
- Spread then moved another 3bp, now down \$4.5mm

Drawdown Framework

Clearly defined intervention point where risk department asks traders to adjust their risk.

- **Protect** firm from losses beyond risk appetite
- **Minimize** negative convexity for traders
 - Why is there potential negative convexity?
- **Forces** conversation between risk and traders, especially when markets are exciting

Clichés

**“Markets can remain irrational longer
than you can remain solvent.”
-John Maynard Keynes**

A Tale of Two Strategies

Level Staking

- Fair Roulette Wheel, Always Bet \$1, Expected Value = 0
- Symmetric Distribution of P&L that if we were continue to spin would approach the Normal distribution

Permutation	Bet	Outcome	Stakes	Profit	Total	Chance	Expectation
1	R,R,R	B,B,B	1,1,1	-1,-1,-1	-3	0.125	-0.375
2	R,R,R	B,B,R	1,1,1	-1,-1,+1	-1	0.125	-0.125
3	R,R,R	B,R,B	1,1,1	-1,+1,-1	-1	0.125	-0.125
4	R,R,R	B,R,R	1,1,1	-1,+1,+1	+1	0.125	+0.125
5	R,R,R	R,B,B	1,1,1	+1,-1,-1	-1	0.125	-0.125
6	R,R,R	R,B,R	1,1,1	+1,-1,+1	+1	0.125	+0.125
7	R,R,R	R,R,B	1,1,1	+1,+1,-1	+1	0.125	+0.25
8	R,R,R	R,R,R	1,1,1	+1,+1,+1	+3	0.125	+0.375

A Tale of Two Strategies

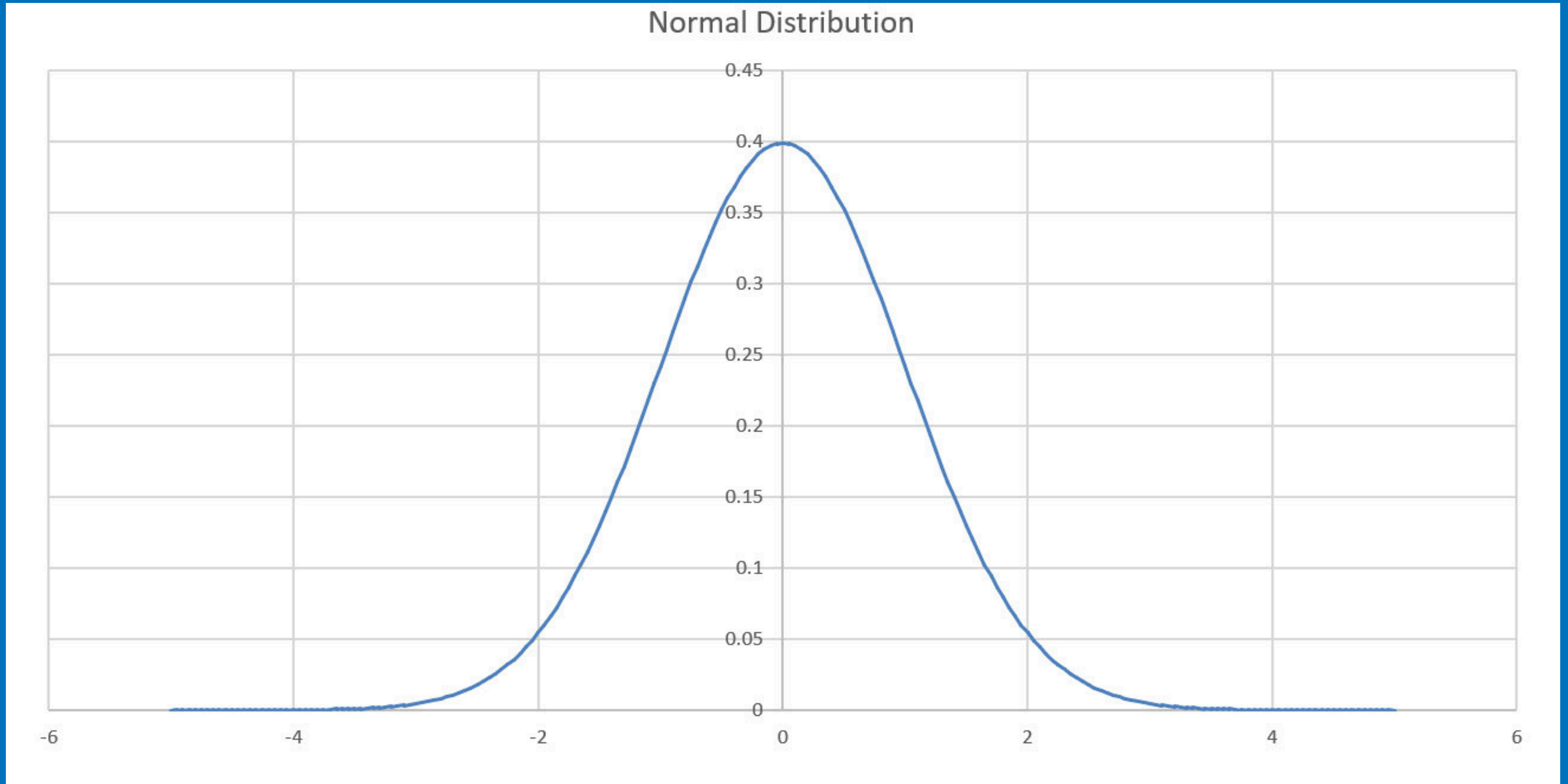
Martingale Strategy

- Double bet size after every loss, Expected value is still 0, Asymmetric P&L distribution, Compared to level betting which lost money $\frac{1}{2}$ the time, here we lose only $\frac{1}{4}$ of the time, but when we lose...

Permutation	Bet	Outcome	Stakes	Profit	Total	Chance	Expectation
1	R,R,R	B,B,B	1,2,4	-1,-2,-4	-7	0.125	-0.875
2	R,R,R	B,B,R	1,2,4	-1,-2,+4	1	0.125	+0.125
3	R,R,R	B,R,B	1,2,1	-1,+2,-1	0	0.125	+0
4	R,R,R	B,R,R	1,2,1	-1,+2,1	2	0.125	+0.25
5	R,R,R	R,B,B	1,1,2	+1,-1,-2	-2	0.125	-0.25
6	R,R,R	R,B,R	1,1,2	+1,-1,+2	2	0.125	+0.25
7	R,R,R	R,R,B	1,1,1	+1,+1,-1	1	0.125	+0.125
8	R,R,R	R,R,R	1,1,1	+1,+1,+1	+3	0.125	+0.25

Normal Distribution

Friend or Foe?



VaR

How much do we lose in X% worst scenario?

CVaR

Conditional on scenarios in the worst X% of simulated days, what is our average loss?

MAD

Average absolute change in value for portfolio for a given time period (usually daily).

VaR and Friends

- **Distributional Measures**

- **VaR**

- The loss level that will not be exceeded with a specified probability
 - Asks the simple question: “***What is a bad day?***”

- **C-VaR (or expected shortfall)**

- The expected loss given that the loss is greater than VaR level
 - Asks the simple question: “***If things do get bad, how much can we expect to lose?***”

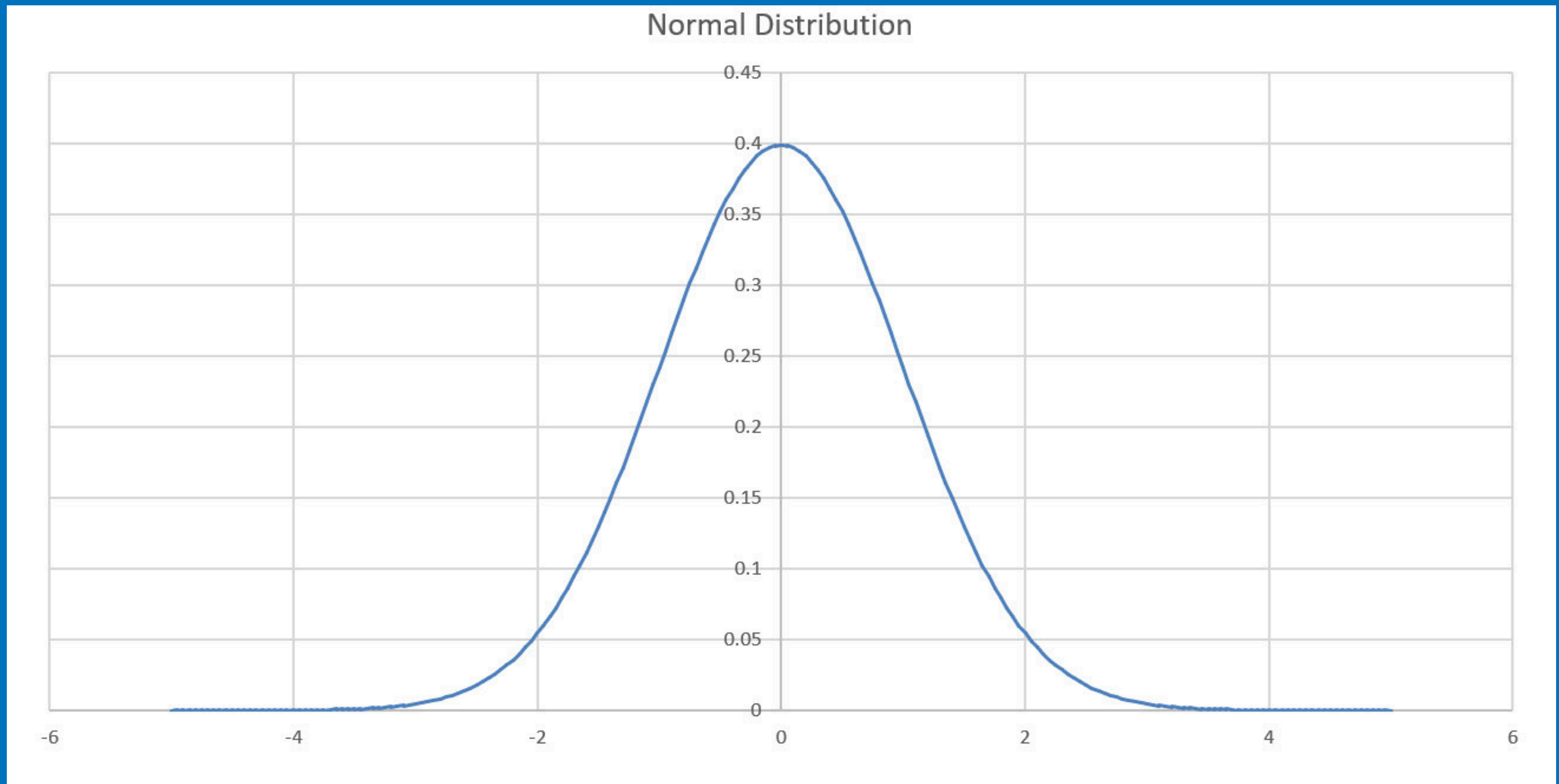
- **MAD**

- Similar to standard deviation
 - More computationally efficient: “***On average, what is the change in value?***”

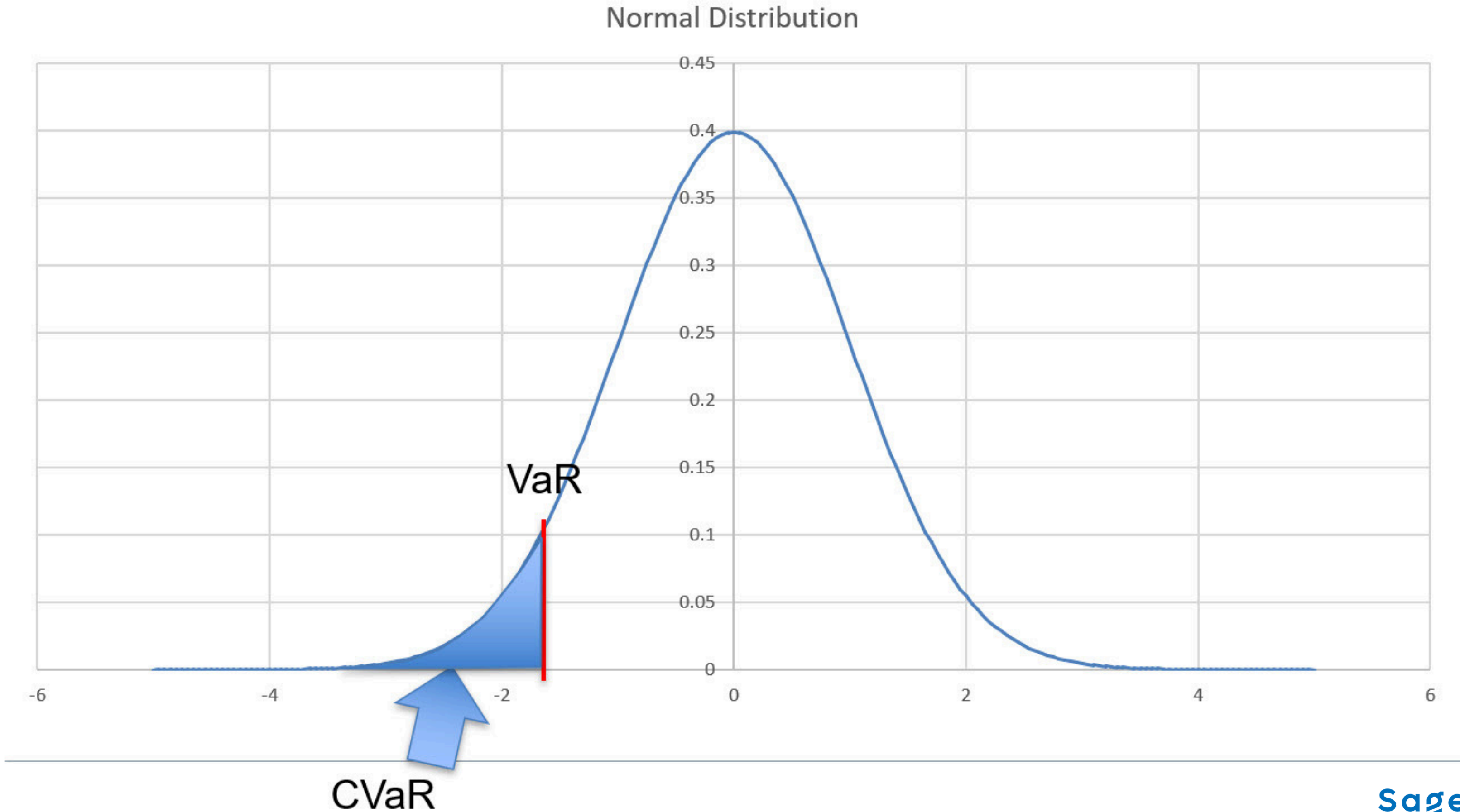
- **Advantages of Distributional Measures**

- Captures an important aspect of risk in a single number
 - Comparable across many asset classes and products (unit is \$)

Draw VaR and CVaR



Draw VaR and CVaR



Downside vs. Upside

- **Distributional Measures**
 - **Sharpe**
 - Ratio between returns and variance:
 - is the annualized return of the portfolio
 - is the annualized return of the risk free rate
 - is the annualized volatility (standard deviation) of the portfolio
- **What is one glaring problem you can see with this measure?**

Downside vs. Upside

- **Distributional Measures**

- **Sortino**

- Adjustment to Sharpe – if upside variance is good, we don't want to penalize ourselves for it
 - Ratio between returns and downside variance:
 - is the annualized downside deviation

- **Disadvantages of these measures**

- “Averages” across many observations, therefore can hide large amounts of tail risk
 - For lower turnover trades, takes lots of observations to have meaning

Calculate VaR, CVaR, MAD, Sharpe, and Sortino

The File

DailyPnL file contains 7 years of single-day historical simulations for a sample portfolio.

Your Task

Calculate the VaR, CVaR, MAD, Sharpe, and Sortino for both buying and selling the portfolio.

Note: For the first two measures, use the 95th percentile.

Extra Credit

For extra credit, calculate the VaR if you scale the standard deviation of this portfolio to the same 95th percentile level?

- How does it compare to the historical measure?
- What concerns do you have with this last part of the exercise?

Stress Testing

- **Define Scenarios, Shock Portfolio**
 - Risk slides standard way to accomplish this
 - Can utilize custom specifications
 - Pick specific unusual dates from the past
- **VaR can be considered a generalization of a stress test**
 - **IMPORTANT** differences:
 - VaR conditions on recent history
 - VaR has opinion about probability
 - If using Gaussian copula, tails are independent

Rush to the Exits...

- **Exchange margins and leverage**
 - When vol goes up, exchanges increase IM to cover their risk
- **Everyone else measured on VaR or some vol-based measure**
 - When vol goes up...per contract risk capacity goes down
- **Drawdown frameworks**
 - When people take losses, risk capacity goes down
- **Known (yet often hidden) accelerants**
 - Delivery of physical assets
 - Highly leveraged positions
 - Crowded positions

Attribution

1st Order:

- **Delta: $\$/d\text{Underlying}$**
 - Delta x Move
- **Theta: $\$/d\text{Time}$**
 - Theta x Time Elapsed
- **Vega: $\$/d\text{Vol}$**
 - Vega x Vol Change

2nd Order:

- **Gamma: $d\text{Delta}/d\text{Underlying}$**
 - $\frac{1}{2} \times \text{Move}^2 * \text{Gamma}$
- **Vega prime: $d\text{Vega}/d\text{Vol}$**
 - $\text{Vol Change}^2 * \text{Vega Prime} / 2$
- **Delta prime: $d\text{Delta}/d\text{Vol}$**
 - $\text{Move} * \text{Vol Change} * \text{Delta Prime}$

Flavors of VaR

95

- Confidence Level
- Percentile Loss

MC

- Stands for Monte Carlo, the simulation type
- “Historical” is the other common alternative

VaR

- Is the measure
- We have also discussed CVaR and MAD

1d1y

- Represents the length of each observation (1day) and the number of days in the dataset (1 year)

.94

- Exponential Weighting
- 50% of variance explained by the most recent 12 days

Quick Review Of Convex Optimization

Definition: A set $C \subseteq \mathbb{R}^n$ is *convex* if for any $x, y \in C$ and any $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \in C$$

Properties:

- ▶ The intersection of convex sets is convex.
- ▶ The convex hull of any set S is the smallest convex set containing S .

Examples:

- ▶ Euclidean space $\mathbb{R}^n$.
- ▶ Convex polyhedra defined by linear inequalities.

Quick Review Of Convex Optimization

Definition: A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* if its domain is a convex set and for any x, y in its domain and $\theta \in [0, 1]$:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

First-order Condition: f is convex if and only if its epigraph is a convex set, or equivalently,

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x), \quad \forall x, y$$

Second-order Condition: If f is twice differentiable, f is convex if and only if its Hessian $\nabla^2 f(x)$ is positive semidefinite for all x in its domain.

Unconstraint Optimization

Problem Statement: Minimize $f(x)$ where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable. **Optimality Condition:**

$$\nabla f(x^*) = 0$$

Interpretation: At the minimum point x^* , the gradient of f is zero, indicating a stationary point.

Constraint Optimization

Problem Statement: Minimize $f(x)$ subject to $x \in C$, where C is a convex set. **Feasible Set:** The set of all x satisfying the

constraints. **Methods:**

- ▶ Projected gradient descent.
- ▶ Lagrange multipliers for equality constraints.

Linear Program

Standard Form:

$$\begin{array}{ll}\min_x & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0\end{array}$$

Properties:

- ▶ Feasible region is a convex polyhedron.
- ▶ Optimal solutions occur at extreme points (vertices).

Linear Program

No Closed-Form Solution for General LP: Linear programming problems generally do not have a closed-form solution due to the constraints $Ax = b$ and $x \geq 0$.

Simplex Method:

- ▶ **Concept:** Simplex is an iterative algorithm that moves along the edges of the feasible polyhedron to find the optimal vertex.
- ▶ **Algorithm Steps:**
 1. **Initialization:** Start at a basic feasible solution (vertex) x_0 .
 2. **Pivoting:** Choose an adjacent vertex that improves the objective value by solving a system of linear equations to update the basic and non-basic variables.
 3. **Termination:** Stop when no adjacent vertex offers further improvement, indicating an optimal solution is reached.

Quadratic Program

Standard Form:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^\top Qx + c^\top x \\ \text{subject to} \quad & Ax \leq b \end{aligned}$$

Properties:

- ▶ If Q is positive semidefinite, the problem is convex.
- ▶ Appears in portfolio optimization and support vector machines.

Quadratic Program

No Closed-Form Solution for General LP: Linear programming problems generally do not have a closed-form solution due to the constraints $Ax = b$ and $x \geq 0$.

Simplex Method:

- ▶ **Concept:** Simplex is an iterative algorithm that moves along the edges of the feasible polyhedron to find the optimal vertex.
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 3. **Termination:** Stop when no adjacent vertex offers further improvement, indicating an optimal solution is reached.

Intro to Lagrangian

Unconstrained Optimization: For an unconstrained optimization problem:

$$\min_x f(x)$$

we simply find the critical points where $\nabla f(x) = 0$. However, real-world problems often include constraints.

Optimization with Constraints: We introduce the Lagrangian function to incorporate the constraints:

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

- ▶ λ_i : Lagrange multipliers for inequality constraints $g_i(x) \leq 0$.
- ▶ ν_j : Lagrange multipliers for equality constraints $h_j(x) = 0$.

Intro to Lagrangian

Bounding Principle: For any feasible point x and any multipliers $\lambda \geq 0, \nu$:

$$f(x) \geq \inf_x \mathcal{L}(x, \lambda, \nu)$$

This inequality shows that the Lagrangian value serves as a lower bound to the primal objective $f(x)$.

Dual Problem: The dual problem seeks to maximize this lower bound:

$$\sup_{\lambda \geq 0, \nu} \inf_x \mathcal{L}(x, \lambda, \nu)$$

- ▶ This bound is crucial because if conditions are right (convexity), the gap between primal and dual solutions vanishes.
- ▶ This means optimizing $\mathcal{L}(x, \lambda, \nu)$ effectively targets the optimal solution within feasible regions.

Intro to Lagrangian

KKT Conditions: Necessary conditions for optimality.

Problem Formulation:

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p \end{aligned}$$

KKT Conditions:

1. **Primal Feasibility:** $g_i(x^*) \leq 0, h_j(x^*) = 0$.
2. **Dual Feasibility:** $\lambda_i^* \geq 0$.
3. **Complementary Slackness:** $\lambda_i^* g_i(x^*) = 0$.
4. **Stationarity:**

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0$$

Intro to Lagrangian

Example: Minimize $f(x) = x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$.
Lagrangian:

$$\mathcal{L}(x, \nu) = x_1^2 + x_2^2 + \nu(x_1 + x_2 - 1)$$

KKT Conditions:

► **Stationarity:**

$$\nabla \mathcal{L} = \begin{bmatrix} 2x_1 + \nu \\ 2x_2 + \nu \end{bmatrix} = 0$$

► **Solve:** From stationarity: $2x_1 + \nu = 0$ and $2x_2 + \nu = 0$.
Combining with $x_1 + x_2 = 1$:

$$x_1 = x_2 = \frac{1}{2}, \quad \nu = -1$$

► **Result:** Optimal solution $x^* = \left(\frac{1}{2}, \frac{1}{2}\right)$.

CLASS EXERCISE

Basic Calculations

Individual Exercise

Data:

- “dataset1.csv”
- Contains 5 return vectors

Your Task:

- Calculate the mean, 95 VaR, 95 CVaR, and MAD
- Answer the questions below:
 1. Are you surprised by any of the results?
 2. Which vector represents the best risk/reward?
 3. Which one is the worst?

Monte Carlo vs. Historical

Individual Exercise

Data:

- “dataset2.csv”

Your Task:

- Refer to actual simulations produced by our VaR system for TYU2021 and a Monte Carlo simulation derived from those.
- Answer the questions below:
 1. Calculate VaR, CVaR, and MAD for each set of vectors
 2. Which one is Hist, which is MC?
 3. Is MC CVaR a useful measure?

Practice Exercise

Class Exercise

Data:

- “unitsims.csv”
- “unitgreeks.csv”
- “position.csv”

Our Task:

- Use the data above to complete the following tasks/questions:
 1. Calculate VaR, CVaR, MAD, Delta, Gamma, Vega, Theta for the existing portfolio.
 2. Using a maximum of 10 new trades, and without adjusting existing options strikes, minimize risk. (We will see how the neutralized portfolio performs over the following five trading days.)

Debrief

Let's review the correct answers
and thought process.

THE END